

TypstRAG Example Paper

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Local RAG over Typst documentation

Abstract. This minimal paper demonstrates the Typst features that TypstRAG retrieves from local Typst documentation: page columns, figures with captions, and bibliography references. The goal is a professor-facing artifact that compiles without a custom template.

Introduction

TypstRAG indexes a pinned Typst documentation tag into a local LanceDB database. The workflow is deliberately small: clone the project, build the index, ask a question, and inspect cited documentation sources.

The example cites prior work on readable markup systems (Knuth, 1984) and local retrieval workflows (Lewis et al., 2020).

Method

The pipeline has four steps: fetch Typst docs, collect source files, split documents into chunks, and build a vector index. A user can then query the index from the CLI or let Hermes use the retrieved context with any current model.

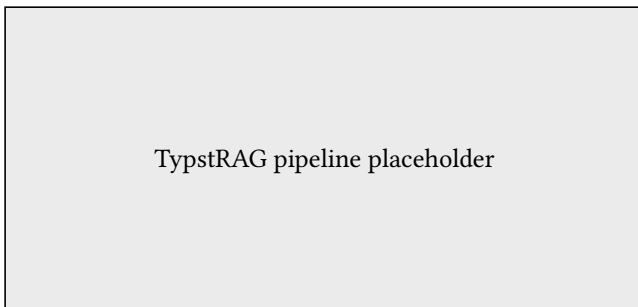


Figure 1: Minimal professor-facing pipeline diagram.

As shown in Figure 1, the artifact is intentionally local-first. This avoids hosting a Python backend just to share a reproducible academic demo.

Result

The current smoke checks verify that the local index opens, search returns rows, and the retrieval evaluation keeps an expected-term hit rate of 1.00 on the small built-in query set.

Conclusion

The project is ready to share as a local RAG package and demo. A future Typst Universe submission should be a separate Typst template package, not the Python RAG itself.

Bibliography

Knuth, D. E. (1984). *The TeXbook*. Addison-Wesley.

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.-tau, Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *Advances in Neural Information Processing Systems*.